MILITARY SPECIFICATION

PROBE, CRUSH SENSOR, RH, FZU-45/B AND LH, FZU-46/B

AND

PROBE, AIR-PRESSURE, FZU-47/B

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of two types of probes, referred to herein as the probe.

1.2 Classification. Probes covered by this specification consist of the following types:

Type I. Probe, Crush Sensor:

a. RH, FZU-45/B (Drawing 642AS5701).

b. LH, FZU-46/B (Drawing 642AS5700).

Type II. Probe, Air-Pressure, FZU-47/B (Drawing 642AS5650).

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1336

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of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-S-6872	Soldering Process, General Specification for.
MIL-T-18303	Test Procedures, Reproduction, Acceptance, and Life for Aircraft Electronic Equipment, Format for.

STANDARDS

Federal

FED-STD-102	Preservation, Packaging, and Packing Levels.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts.
MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Tests for.
MIL-STD-454	Standard General Requirements for Electronic Equipment.
MIL-STD-810	Environmental Test Methods.
MIL-STD-1246	Product Cleanliness Levels and Contamination Control Program.
MIL-STD-1695	Environments, Working, Minimum Standards for.
MIL-STD-45662	Calibration Systems Requirements.

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2.1.2 Other Government documents and drawings. The following other Government documents and drawings of the issue specified in the solicitation, form a part of this specification to the extent specified herein.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 171-178	Transportation.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

Naval Air Systems Command
(Code Ident 30003)

WS-6536	Procedures and Requirements for Preparation and Soldering of Electrical Connections.

DRAWINGS

Naval Air Systems Command
(Code Ident 30003)

642AS5372	Shipping Container Assembly.
642AS5698	Shipping Container Assembly.
SA498014	Fixture, Vibration.
SA498015	Fixture, Vibration.
SA2876068	Fixture, Vibration.
642AS5356	Gas Generator, BBU-32/B.
642AS5548	Clamp Assembly.
642AS5650	Probe, Air-Pressure, FZU-47/B.
642AS5651	Plate, Identification.
642AS5700	Probe, Crush Sensor, LH, FZU-46/B.
642AS5701	Probe, Crush Sensor, RH, FZU-45/B.

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642AS5702	Plate, Identification.
642AS5703	Plate, Identification.

(Copies of specifications, standards, and drawings required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS.

3.1 Item description.

a. Type I. The Type I probe is an electromechanical device which senses target impact and provides a signal to the logic module circuitry upon target impact. (The Type Ia and Type Ib probes are mirror images except for the interface and logic modules.) The major components and general configuration of the Type I probe are shown on Figure 1.

b. Type II. The major components and general configuration of the Type II probe are shown on Figure 2. The Type II probe is an electromechanical device which:

(1) Opens the hermetic seal on the fuze.

(2) Senses dynamic and static air pressure from the missile during free flight, and transfers the resulting pressure to the fuze for arming purposes.

(3) Senses target impact and provides a signal to the logic module circuitry upon target impact.

3.1.1 Government-loaned property. When specified in the contract or purchase order, the Government will loan the contractor and the independent test facility designated in the contract or purchase order, one Fuze and Probe Test Set, Harpoon, referred to herein as the test set (see 6.2.1).

3.1.2 Government-furnished property. When specified in the contract or purchase order, the Government will furnish the independent test facility specified in the contract or purchase order the following support equipment to perform the environmental tests specified herein (see 6.2.1).

a. Fixture, Vibration, Drawing SA498014.

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b. Fixture, Vibration, Drawing SA498015.

c. Fixture, Vibration, Drawing SA2876068.

3.1.3 Autonomous components. When specified in the contract or purchase order, the contractor shall obtain one gas generator in accordance with Drawing 642AS5356 for each probe produced under a specified contract (see 6.2.1).

3.2 First article. When specified, a sample shall be subjected to first article inspection (see 4.4 and 6.2.1).

3.3 Design and construction.

3.3.1 Production drawings. The probe shall meet the requirements of and be manufactured and assembled in accordance with the drawings and documents listed on Drawings 642AS5700 and 642AS5701 for Type I probes, and Drawing 642AS5650 for Type II probes.

3.3.2 Standards of manufacture.

3.3.2.1 Product cleanliness. Product cleanliness shall meet the requirements of MIL-STD-1246, Level 200. No oils or solid lubricants of any kind shall be used on any part of the probe after final cleaning of components, except as specified on the drawings or herein.

3.3.2.2 Working environment. The contractor shall provide facilities for the fabrication, assembly, and testing of the probes to be delivered in accordance with this specification. The working environment shall meet the minimum levels and requirements specified in MIL-STD-1695.

3.3.2.3 Soldering. Procedures and requirements for preparation and soldering of electrical connections shall be in accordance with WS-6536. Mechanical soldering shall be in accordance with MIL-S-6872.

3.4 Performance characteristics.

3.4.1 Probe erection.

3.4.1.1 Type I. The probe shall erect to a height of 2.75 ±0.25 inches and shall meet the requirements of 3.4.2.1 and 3.4.3 (see 4.7.2.1).

3.4.1.2 Type II. The probe shall erect to a height of 2.75 ±0.25 inches. The differential leak check of the air passages shall not show more than 4 inches of water pressure differential when input air pressure is equal to 60 ±1 inches of water. The probe shall meet the resistance requirements of Table IV, Tests A2, A3, and A4 (see 4.7.2.2).

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3.4.2 Electrical. The following requirements shall be met by the unpotted (Type II only), but welded probe mechanism assembly, and by the completed end item. An electrical diagram of the probe is shown on Figure 3 (Type Ia), Figure 4 (Type Ib), and Figure 5 (Type II).

3.4.2.1 Resistance (Type I).

3.4.2.1.1 Type Ia. The Type Ia probe shall meet the requirements of Table I (see 4.7.3.1.1).

TABLE I. Probe resistance requirements, type Ia.

Test	Selector switch 1/	Resistance parameters tested	Lower limit	Upper limit
Mode HP 2/				
R1	1	M1, M2, K1, K2 to case	100 Mohms	...
R2	2	F1 to case	100 Mohms	...
Mode 3469				
R3 3/	3	B1 to case	...	0.5 ohm
R4	4	F1 to E1	...	0.5 ohm
R5	5	M1 to M2	0.8 ohm	1.6 ohms
R6	6	M1 to K2	1.6 ohms	3.2 ohms
R7	7	M1 to K1	0.8 ohm	1.6 ohms
R8	8	M2 to K2	2.4 ohms	4.8 ohms
R9	9	K1 to M2	1.6 ohms	3.2 ohms
R10	10	K1 to K2	0.8 ohm	1.6 ohms
R11 4/	NA	B2 to P2	0.0 ohm	0.5 ohm

1/ Selector switch position number refers to the Fuze and Probe Test Set, Harpoon.

2/ The High Pot (HP) tests shall be performed at 500 ±25 volts direct current (Vdc).

3/ Shorting clip between T10 on test set and case of probe.

4/ Measured at TP1 and TP2 of test set.

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3.4.2.1.2 Type Ib. The Type Ib probe shall meet the requirements of Table II (see 4.7.3.1.2).

TABLE II. Probe resistance requirements, type Ib.

Test	Selector switch[1/]	Resistance parameters tested	Lower limit	Upper limit
Mode HP[2/]				
R1	1	K1, K2, L1, L2 to case	100 Mohms	...
R2	2	A2 to case	100 Mohms	...
Mode 3469				
R3[3/]	3	B1 to case	...	0.5 ohm
R4	4	A1 to A2	...	0.5 ohm
R5	5	L2 to L1	0.8 ohm	1.6 ohms
R6	6	L2 to K1	1.6 ohms	3.2 ohms
R7	7	L2 to K2	0.8 ohm	1.6 ohms
R8	8	L1 to K1	2.4 ohms	4.8 ohms
R9	9	L1 to K2	1.6 ohms	3.2 ohms
R10	10	K2 to K1	0.8 ohm	1.6 ohms
R11[4/]	NA	H1, 2, M1, 2, J1, 2, P1, 2	4.75 kohms	5.25 kohms

1/ Selector switch position number refers to the Fuze and Probe Test Set, Harpoon.

2/ The HP tests shall be performed at 500 ±25 Vdc.

3/ Shorting clip between T10 on test set and case of probe.

4/ Measured at TP1 and TP2 of test set.

3.4.2.2 Resistance (Type II).

3.4:2.2.1 Safe condition. With the probe mechanism assembly in the safe condition (see 6.3.1), the probe shall meet the resistance requirement of Table III (see 4.7.3.2.1).

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TABLE III. Probe resistance requirements, type II (safe condition).

Test	Selector switch 1/	Resistance parameters tested	Lower limit	Upper limit
Mode HP 2/				
S1	1	T4, T5, T7, T8 to case	100 Mohms	...
S2	2	T2 to case	100 Mohms	...
Mode 3469				
S3 3/	3	T6, T9 to case	...	0.5 ohm
S4	4	T1 to T2	...	0.5 ohm
S5	5	T4 to T5	1.7 kohms	2.3 kohms
S6	6	T5 to T8	3.4 kohms	4.6 kohms
S7	7	T5 to T7	1.7 kohms	2.3 kohms
S8	8	T4 to T8	1.7 kohms	2.3 kohms
S9	9	T4 to T7	0.8 ohm	1.6 ohms
S10	10	T7 to T8	1.7 kohms	2.3 kohms

1/ Selector switch position number refers to the Fuze and Probe Test Set, Harpoon.

2/ The HP tests shall be performed at 500 ±25 Vdc.

3/ Shorting clip between T10 on test set and case of probe.

3.4.2.2.2 Armed condition. With the probe mechanism assembly in the armed condition (see 6.3.2), the probe shall meet the resistance requirements of Table IV (see 4.7.3.2.2).

3.4.3 Operational (Type I only).

3.4.3.1 Logic module operational check (Type Ia). The logic module of the Type Ia probe shall meet the requirements of Table V (see 4.7.4.1).

3.4.3.2 Interface module operational check (Type Ib). The interface module of the Type Ib probe shall meet the requirements of Table VI (see 4.7.4.2).

3.4.4 Environmental conditions. The probe shall not be damaged nor shall operability, safety, and reliability performance be degraded by exposure to the environments specified in 3.4.4.1 through 3.4.4.5.

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TABLE IV. Probe resistance requirements, type II (armed condition).

Test	Selector switch 1/	Resistance parameters tested	Lower limit	Upper limit
Mode HP 2/				
A1	1	T4, T5, T7, T8 to case	100 Mohms	...
A2	2	T2 to case	100 Mohms	...
Mode 3469				
A3 3/	3	T6, T9 to case	...	0.5 ohm
A4	4	T1 to T2	...	0.5 ohm
A5	5	T4 to T5	0.8 ohm	1.6 ohms
A6	6	T5 to T8	2.4 ohms	4.8 ohms
A7	7	T5 to T7	1.6 ohms	3.2 ohms
A8	8	T4 to T8	1.6 ohms	3.2 ohms
A9	9	T4 to T7	0.8 ohm	1.6 ohms
A10	10	T7 to T8	0.8 ohm	1.6 ohms

1/ Selector switch position number refers to the Fuze and Probe Test Set, Harpoon.

2/ The HP tests shall be performed at 500 ±25 Vdc.

3/ Shorting clip between T10 on test set and case of probe.

3.4.4.1 Temperature and humidity. The probe shall be safe (see 6.3.3) and operable (see 6.3.4) following exposure to the environment of 4.7.6.1.

3.4.4.2 Transportation vibration-temperature. The probe shall be safe (see 6.3.3) and operable (see 6.3.4) following exposure to the environment of 4.7.6.2.

3.4.4.3 Shock.

3.4.4.3.1 Handling shock. The probe shall be safe (see 6.3.3) and operable (see 6.3.4) following exposure to the environment of 4.7.6.3.1.

3.4.4.3.2 Pyro shock. The probe shall be safe (see 6.3.3) and operable (see 6.3.4) following exposure to the environment of 4.7.6.3.2.

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TABLE V. Logic module operational check, type Ia.

Test	Switch 1	Switch 2	Switch 3	Switch 4 (reset)	Switch 5	Switch 6	Switch 7	Test point 2 output voltage
Operational requirements								
1 1/	0	0	0	0	0	0	0	<1 volt
2	X	0	0	0	0	0	0	<1 volt
3	0	X	0	0	0	0	0	>10 volts
4	0	0	0	X	0	0	0	<1 volt
5	0	0	X	0	0	0	0	<1 volt
6	X	0	0	0	0	0	0	>10 volts
7	0	0	0	X	0	0	0	<1 volt
8	0	X	0	0	0	0	0	<1 volt
9	0	0	X	0	0	0	0	>10 volts
10	0	0	0	X	0	0	0	<1 volt
11	X	X	X	0	0	0	0	>10 volts
12	0	0	0	X	0	0	0	<1 volt
13	0	0	0	0	X	0	0	<1 volt
14	0	0	0	0	0	X	0	>10 volts
15	0	0	0	X	0	0	0	<1 volt
16	0	0	0	0	0	0	X	<1 volt
17	0	0	0	0	X	0	0	>10 volts
18	0	0	0	X	0	0	0	<1 volt
19	0	0	0	0	0	X	0	<1 volt
20	0	0	0	0	0	0	X	>10 volts
21	0	0	0	0	X	0	0	>10 volts
22	The voltage at test point 1 shall be 12 ± 1.5 Vdc							

Legend:

0 = Normal.
X = Momentarily depressed.

NOTE:

1/ Initialize unit by depressing switch 4 (reset).

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TABLE VI. Interface module operational check, type Ib.

Test	Switch position			Test point voltage 1/	
	S1	S2	S3	TP1	TP2
1	O	O	O	0.3 V max	10 V min
2	O	X	O	10 V min	10 V min
3	X	O	O	10 V min	1 V max
4	X	O	X	0.3 V max	10 V min
5	X	X	X	10 V min	10 V min
6	TP3 to TP4			1.0 ohm max	

Legend

O = Normal.
X = Depressed.

NOTE:

1/ Shall not exceed 15 Vdc.

3.4.4.3.3 Thermal shock. The probe shall remain safe (see 6.3.3) during the test and be operable (see 6.3.4) after exposure to the environment of 4.7.6.3.3.

3.4.4.4 Captive flight and boost flight vibration (nonerected). The probe shall be safe (see 6.3.3) and operable (see 6.3.4) following exposure to the environment of 4.7.6.4.

3.4.4.5 Missile flight vibration (erected). The probe shall meet the following requirements following exposure to the environment of 4.7.6.5.

a. Type I. The probe shall remain erected and shall meet the requirements of 3.4.1.1.

b. Type II. The probe shall remain erected and shall meet the requirements of 3.4.1.2.

3.5 Visual. The safety plate (Type I) and safety plate assembly (Type II) shall be in place. The probe shall have no obvious damage or defects which could affect operability, safety, or reliability (see 4.7.5).

3.6 Marking. Marking of the probe shall be in accordance with Drawing 642AS5703 (Type Ia), Drawing 642AS5702 (Type Ib) and Drawing 642AS5651 (Type II) (see 4.7.7).

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3.7 Workmanship. The probe and all components shall meet the workmanship requirements of MIL-STD-454, Requirement 9 (see 4.7.7).

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein (see 6.2.1). Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government (see 6.2.1). The Government reserves the right to perform or witness any of the inspections set forth in this specification where such inspections are deemed necessary to ensure that supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. The contractor shall ensure that test and inspection facilities of sufficient accuracy, quality, and quantity are established and maintained to permit performance of required inspections.

4.1.1.1 Test equipment calibration. All test equipment shall be calibrated and maintained in accordance with the requirements of MIL-STD-45662. When specified in the contract or purchase order, the contractor shall prepare a calibration plan in accordance with the Data Item Description (DID) cited in 6.2.2. The calibration plan shall be acceptable to the Government prior to use.

4.1.1.2 Test equipment accuracy. Unless otherwise specified herein, all test equipment used in the performance of the specified tests shall have an accuracy of at least 10 percent of the tolerance for the variables to be measured.

4.1.1.3 Special test equipment and circuits. When special test equipment and circuits, other than as specified herein or supplied by the Government (see 3.1.1), are devised or commercially-available equipment is employed, all test equipment, circuits, and methods shall meet the requirements of 4.1.1.1 and 4.1.1.2 and are subject to acceptance by the procuring activity.

4.1.1.4 Gages. The contractor shall provide whatever gages are necessary to ensure conformance with the dimensional requirements of this specification and the applicable drawings. Unless otherwise specified in the contract or purchase order, the procuring activity will furnish drawings of pertinent Government final inspection gages (see 6.2.1). This procedure does not relieve the contractor of his responsibility for the design and manufacture of the gages required for satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Government final acceptance gages.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspections (see 4.4).

b. Quality conformance inspections (see 4.5).

c. Individual unit acceptance inspections (see 4.6).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in 3.3.2.2.

4.3.1 Test plan and test procedures. When specified in the contract or purchase order, the contractor shall prepare, using MIL-T-18303 as a guide, a test plan and test procedure in accordance with the DID cited in 6.2.2. The test plan and test procedure shall be acceptable to the Government prior to use.

4.3.2 Lot inspection report. When specified in the contract or purchase order, the contractor shall prepare a lot inspection report for each lot produced under a specified contract, including first article, inspection, and rejected lots, in accordance with the DID cited in 6.2.2. This report shall contain the data necessary for the technical monitoring of lot inspection.

4.3.3 Production summary report. When specified in the contract or purchase order, the contractor shall prepare a production summary report in accordance with the DID cited in 6.2.2. The report shall contain the data necessary for monitoring production by the Government.

4.3.4 Inspection record. When specified in the contract or purchase order, the contractor shall prepare an inspection record in accordance with the DID cited in 6.2.2 for each probe produced under a specified contract. The required test data shall be extracted from the inspection and acceptance records maintained by the contractor.

4.3.5 Inspection and acceptance records. Inspection and acceptance records shall be maintained by the contractor and shall be organized to facilitate search and review. These records shall include the results of all tests and inspections specified herein and any additional tests, inspections, repairs, rework, replacements, or retests of any part, subassembly, assembly, or end-item deemed necessary by the contractor to achieve acceptance. These records shall be made available to the procuring activity on request.

4.4 First article inspections. When specified in the contract or purchase order, a first article sample of 20 probes each of Type Ia, Ib, or II shall be delivered to an independent testing activity designated for first article testing (see 6.2.1). The first article sample shall be

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manufactured using the same methods, materials, equipment, and processes proposed for production. First article approval is valid only on the contract under which it is granted, unless extended by the Government to other contracts. Any production prior to approval of the first article shall be at the risk of the contractor.

4.4.1 First article inspection sequence. The first article sample shall be subjected to the individual unit acceptance inspections of Figure 6 (see 4.6). The first article shall then be subjected to the inspections of Figures 7 and 8 in the sequence shown. The inspections of Figure 8 shall be performed at an independent facility designated in the contract or purchase order (see 6.2.1).

4.4.2 First article acceptance and rejection criteria. When subjected to the inspection sequences of Figures 7 and 8, each probe of the first article sample shall meet the applicable performance requirements of Section 3 on the first and only try. Defects are classified as CRITICAL or MAJOR, as defined in MIL-STD-105, and are specified herein and on the drawings. The following shall be cause for refusal to grant first article approval:

a. One or more CRITICAL defects are found, or

b. Two or more MAJOR defects are found.

4.5 Quality conformance inspections.

4.5.1 Inspection lot. Unless otherwise specified in the contract or purchase order, an inspection lot shall contain one type only and not less than 51 nor more than 280 including those required for destructive testing (see 6.2.1). An inspection lot, as defined in MIL-STD-105, shall consist of completed probes fabricated under the same conditions and offered for inspection at the same time.

4.5.1.1 Serial numbers. Each probe of an inspection lot shall be assigned a Government serial number in order of consecutive assembly, as specified in the contract or purchase order (see 6.2.1). After assignment of a Government serial number to each probe, the inspection lot shall be formed and identified by lot number.

4.5.2 Quality conformance inspection sequence. Prior to formation of the lot, the probe shall be subjected to the individual unit acceptance inspections of Figure 6 (see 4.6). After the inspection lot has been formed and identified by number, the lot shall be subjected to the inspections of Figure 9. After completion of the inspections of Figure 9, nine probes each of one type only shall be randomly sampled from the inspection lot in accordance with MIL-STD-105 and subjected to the inspections of Figure 8 at an independent facility designated in the contract or purchase order (see 6.2.1). The remainder of the lot shall be held in bond pending acceptance of the 9-probe sample.

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4.5.3 Quality conformance acceptance and rejection criteria. Defects in the probes are classified as CRITICAL or MAJOR, as defined in MIL-STD-105, and are specified herein and on the drawings. Each probe of the inspection lot shall meet the performance requirements of Section 3 on the first and only try. For the inspections of Figure 9, acceptance or rejection shall be on an individual basis, except where sampling (see 4.5.3.1) is indicated. For the inspections of Figure 8, one or more CRITICAL or MAJOR defectives shall cause rejection of the inspection lot.

4.5.3.1 Sampling. For inspection lot sampling (see Figure 9), the following criteria shall be used for lot acceptability:

a. For lot sizes of 51 to 159 probes, lot acceptability shall be determined in accordance with MIL-STD-105, Single Sampling Plan, Tightened Inspection, Sample Size Code Letter H, Table II-B, acceptable quality level (AQL) of 0.40 percent defective.

b. For lot sizes at 160 to 280 probes, lot acceptability shall be determined in accordance with MIL-STD-105, Double Sampling Plan, Normal Inspection, Sample Size Code Letter K, Table III-A, AQL of 0.40 percent defective.

4.5.3.2 Counting of defects. When counting defects found in the probes to determine acceptability of the lot, probes with multiple defects shall be counted as a single defect, using the highest classification of the defects found. When a probe has both a CRITICAL and MAJOR defect, it shall be counted as a CRITICAL defect.

4.6 Individual unit acceptance inspections. Individual unit acceptance inspections shall be performed during the manufacture of the probe in accordance with Figure 6. Parts, subassemblies, or assemblies found to be defective during the inspection sequence shall be rejected.

4.7 Methods of inspection. The inspections specified in 4.7.1 through 4.7.8 shall be performed to determine conformance to the applicable requirements of Section 3.

NOTE: The results of all inspections shall be recorded on the inspection record (see 4.3.2).

4.7.1 Drawing inspections. To determine conformance to 3.3.1, perform drawing inspections on each lot of parts, subassemblies, and assemblies of the probe during manufacture, and on the completely assembled probes in accordance with Drawings 642AS5700 and 642AS5701 for Type I probes, and 642AS5650 for Type II probes.

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4.7.2 Probe erection.

4.7.2.1 Type I. To determine conformance to 3.4.1.1, perform the following erection test sequence:

a. Mount the probe in a firing chamber so the probe can erect to a clear height of 2.85 inches.

b. Connect a firing box capable of delivering 10 amperes (A) at 28 Vdc to the proper pins on the module connectors of the probe being tested.

(1) Type Ia. Connect the positive side of the firing box to M2 and K2 and the ground side to M1 and K1.

(2) Type Ib. Connect the positive side of the firing box to L1 and K1 and the ground side to L2 and K2.

c. Close door of firing chamber and discharge firing box into the probe.

d. Remove the probe from the firing chamber.

e. Perform the following inspections:

(1) Type Ia. Perform the inspections of 4.7.3.1.1 (tests R2, R3, and R4 only) and 4.7.4.1.

(2) Type Ib. Perform the inspections of 4.7.3.1.2 (tests R2, R3, and R4 only) and 4.7.4.2.

M101

4.7.2.2 Type II. To determine conformance to 3.4.1.2, perform the following erection test sequence:

a. Install the simulated air tube manifolds (Figure 10) into the probe using the air tube clamp shown on Drawing 642AS5548.

b. Mount the probe in a firing chamber so the probe can erect to a clear height of 2.85 inches.

c. Connect a firing box capable of delivering 10 A at 28 Vdc. To provide the two inputs to the firing box, parallel the two 22-gage white wires for one input and the two 22-gage blue wires for the other input. Remove the safety plate assembly.

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d. Close door of firing chamber and discharge firing box into the probe.

e. Remove the probe from the firing chamber. Connect the probe to the test set (or equal test equipment) using the diagram of Figure 11.

f. Verify each air path separately using 60 ±1 inches of water pressure input (see 3.4.1.2).

M102

g. Perform resistance checks A2, A3, and A4 of Table IV.

M103

4.7.3 Electrical. Resistance measurements shall be performed using the test set (see 3.1.1), or standard test equipment that meets the requirements of 4.1.1.3. All leads shall remain connected during the tests of 4.7.3.1 and 4.7.3.2 when using the test set. Resistance readings of less than 5 ohms shall be adjusted to compensate for internal test equipment resistance.

WARNING

The probe contains pyrotechnics. Proper safety precautions must be observed (see 6.4).

4.7.3.1. Resistance (Type I).

4.7.3.1.1 Type Ia. Verify the Type Ia probe meets the requirements of 3.4.2.1.1 (see Figure 3 and Table I).

M104

4.7.3.1.2 Type Ib. Verify the Type Ib probe meets the requirements of 3.4.2.1.2 (see Figure 4 and Table II).

M105

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4.7.3.2 Resistance (Type II).

4.7.3.2.1 Safe condition. With the probe in the safe condition (see 6.3.1) and the safety wire of the safety plate assembly through the switch plunger, verify the resistance requirements of 3.4.2.2.1 (see Figure 5 and Table III).

M106

4.7.3.2.2 Armed condition. With the probe in the armed condition (see 6.3.2) and the safety wire of the safety plate assembly removed from the switch plunger, verify the resistance requirements of 3.4.2.2.2 (see Figure 5 and Table IV).

M107

4.7.4 Operational (Type I only).

4.7.4.1 Logic module operational check (Type Ia). With the Type Ia module connected to the logic module test circuit (see Figure 12), perform the test sequence of Table V while monitoring the output voltage at test point 2. The module shall meet the requirements of 3.4.3.1.

M108

4.7.4.2 Interface module operational check (Type Ib). With the Type Ib module connected to the interface module test circuit (see Figure 13), perform the test sequence of Table VI while monitoring the voltage at test points 1 and 2. The resistance between test points 3 and 4 shall be less than 1.0 ohm. The module shall meet the requirements of 3.4.3.2.

M109

4.7.5 Visual inspection. To determine conformance to 3.5, each probe shall be visually inspected for:

a. Any external damage that may indicate impairment to operability, safety, or reliability of the probe.

M110

b. Safety plate assembly correctly installed (Type I and II), and safety wire through the switch plunger (Type II).

C1

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4.7.6 Environmental. All probes of the first article (see 4.4) and a randomly selected sample from each inspection lot (see 4.5.2) shall be subjected to the environmental tests in the sequence shown on Figure 8 at a facility designated in the contract or purchase order (see 6.2.1). Any probe subjected to these tests shall be considered destroyed and shall not be returned to the lot nor forwarded for service use. The probe shall be labeled appropriately and held for disposal instructions from the procuring activity (see 6.2.1). Initiation of the pyrotechnic component, except when activated intentionally as part of a test, shall be considered a MAJOR defect and shall cause rejection of the lot from which the probe was drawn.

WARNING

The probe contains pyrotechnics. Proper safety precautions must be observed (see 6.4).

4.7.6.1 Temperature and humidity. To determine conformance to 3.4.4.1, test the probe in accordance with MIL-STD-331, Test 105.1. Protected covers shall be used on Type I connectors for this test.

Remain safe:	C2
Remain operable:	M111

4.7.6.2 Transportation vibration-temperature. To determine conformance to 3.4.4.2, test the probe in accordance with MIL-STD-331, Test 119, Procedure 2.

Remain safe:	C3
Remain operable:	M112

NOTE: For the tests of 4.7.6.2 through 4.7.6.6, the probe gasket (Figure 14, Type I; Figure 15, Type II) shall be mounted between the probe flange and the vibration fixture. The probe and gasket shall be mounted to the vibration fixture (Drawings SA2876068 Type I, SA498014 or SA498015 Type II), and the fixture then mounted to the vibration table with the mounting bolts tightened to a torque of 25 ±1 foot-pounds. Screws for mounting Type I probes shall be tightened to a torque of 20 to 25 inch-pounds. Screws for mounting Type II probes shall be tightened to a torque of 12 to 15 inch-pounds.

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4.7.6.3 Shock.

4.7.6.3.1 Handling shock. To determine conformance to 3.4.4.3.1, test the probe in accordance with MIL-STD-202, Method 213, Test Condition J, except the amplitude shall be 15 gravity units (g).

Remain safe:	C4
Remain operable:	M113

4.7.6.3.2 Pyro shock. To determine conformance to 3.4.4.3.2, test the probe in accordance with the following. Subject the probe to one transient oscillation in the primary axis (see Figure 1 and 2) with the shock response spectrum shown in Figure 16, and one transient oscillation in each of the two remaining orthogonal secondary axes with the shock response spectrum shown in Figure 16. The duration of the applied shock transients shall be between 0.05 and 0.5 second. The analyses used to determine the shock response spectrum shall be performed at a sequence of natural frequencies at intervals of one sixth octave or smaller. The allowed tolerance will be ±6 dB from the specified spectrum except for any given range of frequencies equal to one octave, the test levels shall be within ±3 dB of the specified frequency range for at least one half of the octave.

Remain safe:	C5
Remain operable:	M114

4.7.6.3.3 Thermal shock. To determine conformance to 3.4.4.3.3, test the probe in accordance with MIL-STD-810, Method 503.1, 2 cycles.

Remain safe:	C6
Remain operable:	M115

4.7.6.4 Captive flight and boost flight vibration (nonerected). To determine conformance to 3.4.4.4, test the probe in accordance with MIL-STD-810, Method 514, Procedure V, Part 2. For the captive flight vibration portion use curve shown on Figure 17 herein, except vibration shall be applied for a period of 2 hours along each at the three mutually perpendicular axes for a total of 6 hours. For the boost flight vibration portion use curve shown on Figure 18 herein, except vibration shall be applied for a period of 1 minute minimum along each of three mutually perpendicular axes for a total of 3 minutes.

Remain safe:	C7
Remain operable:	M116

4.7.6.5 Missile flight vibration (erected). To determine conformance to 3.4.4.5, test the probe in the erected position in accordance with MIL-STD-810, Method 514, Procedure V, Part 2, Curve shown on Figure 17 herein. Random vibration shall be applied for 10 minutes minimum along each of the three mutually perpendicular axes for a total

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of 30 minutes minimum. The outputs of the probes will be monitored during the test and shall meet the following requirements:

a. Type Ia probe. Perform the following test sequence through the mating connector to the electronics module:

(1) Connect pin A1 to pin A2.

(2) Connect pin F1 to pin F2.

(3) Connect pin E1 to pin E2.

(4) Supply 26 ±4 Vdc to pins H2(+) and H1(-).

(5) Momentarily connect pin G1 to H1 (reset).

(6) Monitor between pins G2(+) and H1(-) with high impedance voltmeter. The monitored voltage shall remain below 1 Vdc prior to and during the test.

b. Type Ib probe. Perform the following test sequence through the mating connector to the electronics module:

(1) Connect pin C2 to pin F2.

(2) Supply 26 ±4 Vdc to pins B2(+) and G1(-).

(3) Supply 10 ±1 Vdc to pins E1(+) and G1(-).

(4) Monitor with a latching device to detect continuity between pins A1 and B1. No closure shall occur between pins A1 and B1 during the test.

(5) Monitor between pins D1(+) and G1(-) with a high impedance voltmeter. The voltage shall remain greater than 10 Vdc prior to and during the test.

(6) Monitor between pins C1(+) and G1(-) with a high impedance voltmeter. The voltage shall remain greater than 10 Vdc prior to and during the test.

c. Type II probe. Monitor with a latching device to detect continuity between terminals T1 and T3. No closure shall occur between terminals T1 and T3 during the test.

Electrical inspections: M117 Remain erected: M118 Differential leak check of 3.2.1.2: M119

4.7.7 Final visual. Each probe shall be visually examined to determine conformance to 3.6 and 3.7.

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4.7.8 Packaging inspection. Prior to shipment, inspect the packaging, packing, and marking of the probe to determine conformance to Section 5.

5. PACKAGING.

5.1 Preservation-packaging. Preservation and packaging shall be Level C in accordance with FED-STD-102 and as specified on Drawing 642AS5698 for Type I probes, or Drawing 642AS5372 for Type II probes.

5.1.1 Level C. Probes shall be packaged in accordance with Drawing 642AS5698 for Type I probes, or Drawing 642AS5372 for Type II probes. Four probes shall be packaged in each container in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during direct shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be Level C in accordance with FED-STD-102.

5.2.1 Level C. Probes, packaged in accordance with 5.1.1, shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. Packing and containers shall conform to 49 CFR 171-178.

5.3 Marking. In addition to any special marking required by the contract or purchase order, marking shall be in accordance with Drawing 642AS5698 for Type I probes or Drawing 642AS5372 for Type II probes and MIL-STD-129 (see 6.2.1).

6. NOTES.

6.1 Intended use. The probe is intended for use in the fuzing system of a Harpoon air-, surface-, or subsurface-launched guided missile to provide arming energy to the fuze and to sense target impact.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Number, title, and date of this specification.

b. Government-loaned property. The contracting officer should arrange to loan the property cited in 3.1.1.

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c. Government-furnished property. The contracting officer should arrange to furnish the property cited in 3.1.2.

d. Autonomous components. The contracting officer should arrange to furnish the component cited in 3.1.3.

e. First article. When a first article is required, it should be tested and approved under the appropriate provisions of 7-104-55 of the Defense Acquisition Regulations (DAR). The first article should be a preproduction sample consisting of 20 probes each of Type Ia, Ib, or II as specified in 4.4. The contracting officer should include specific instructions in all instruments regarding arrangements for examinations, tests, approval, and disposition of the first article.

f. Responsibility for inspection and inspection facilities, if other than as specified in 4.1.

g. Independent test facility to perform the inspections of Figure 8 (see 4.4, 4.5.2, and 4.7.6).

h. Inspection lot size, if other than as specified in 4.5.1.

i. Assignment of serial numbers (see 4.5.1.1).

j. Government final inspection gage drawings, if required (see 4.1.1.4).

k. Disposal of probes subjected to environmental tests (see 4.7.6).

l. Special markings, if required (see 5.3).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below should be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated in the contract. When the provisions of DAR 7-104.9(n)(2) are invoked and the DD Form 1423 is not used, the data specified below should be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
4.1.1.1	Calibration Plan	DI-R-7064
4.3.1	Test Plans and Test Procedures	DI-T-5204
4.3.2	Lot Inspection Report	UDI-R-20344B
4.3.3	Production Summary Report	DI-A-5030B
4.3.4	Inspection Record	UDI-M-20404

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(Copies of DIDs required by the contractor in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Definitions.

6.3.1 Safe condition.

6.3.1.1 Type I. The Type I probe is considered to be in the safe condition when the safety plate is properly installed.

6.3.1.2 Type II. The Type II probe is considered to be in the safe condition when the safety wire of the safety plate assembly is inserted in the probe switch.

6.3.2 Armed condition (Type II). The probe is considered to be in the armed condition when the safety wire of the safety plate assembly is removed from the probe switch.

6.3.3 Safe. Safe is defined as the ability of the probe to maintain its safety features in a condition which will not create a hazard for personnel or cause any subsequent action to occur during its service life which will nullify the required safety conditions of service use. For nonservice use, safe is defined as the ability of the probe to maintain its safety features in a condition which will permit disposal without injury to personnel.

6.3.4 Operable. Before probe erection, operable is defined as the ability of the probe to perform to completion its specified function and sequence. Upon probe erection, operable is defined as the ability of the probe to produce the specified outputs within the required operational period at the specified point of time.

6.4 Explosives safety precautions. Minimum explosives safety precautions for use by the contractor are detailed in DoD Instruction 4145.26M, DoD Contractors' Safety Manual for Ammunition, Explosives, and Related Dangerous Material.

Preparing activity
Navy-AS
(Project 1336-N429)

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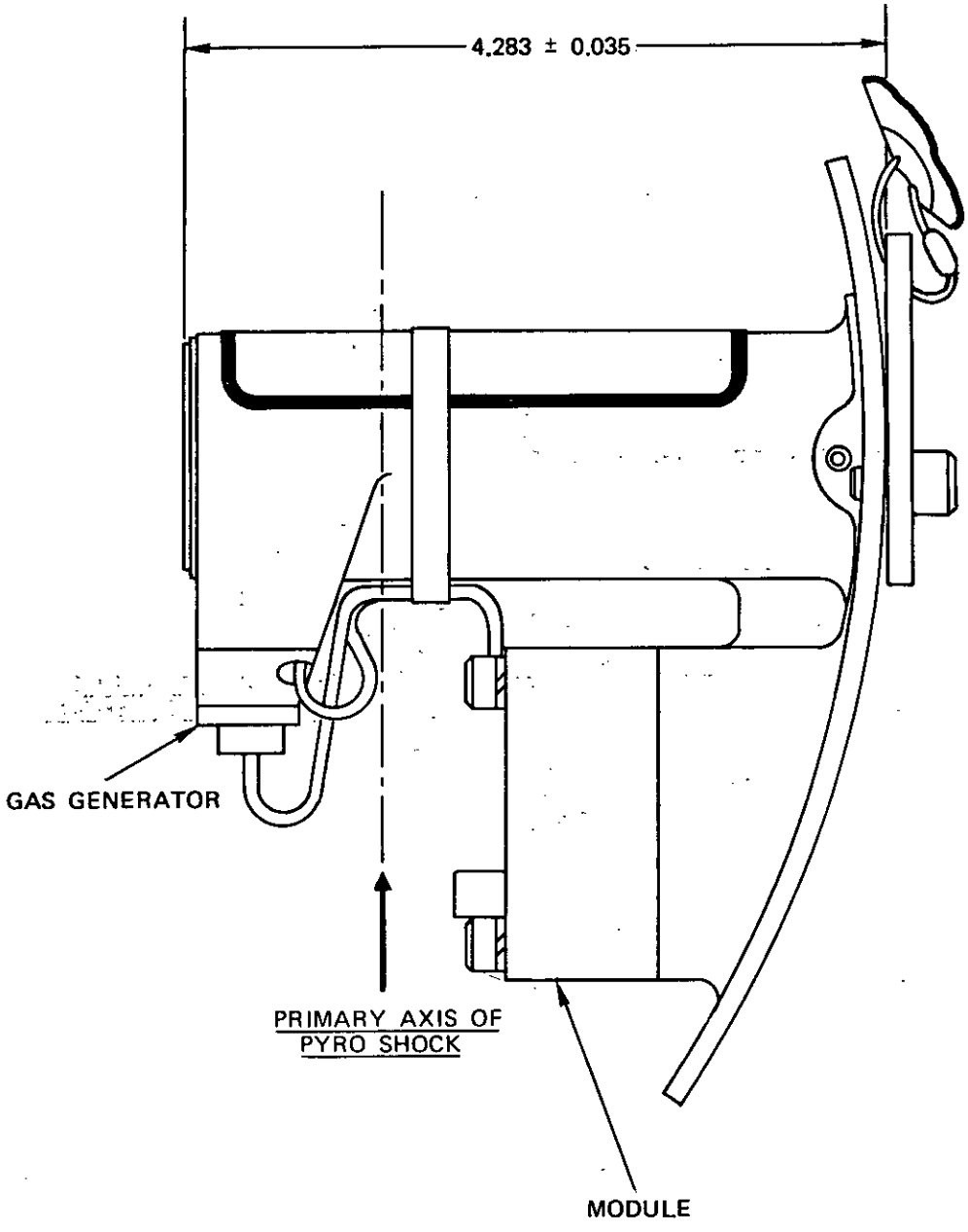


FIGURE 1. Probe configuration, type I.

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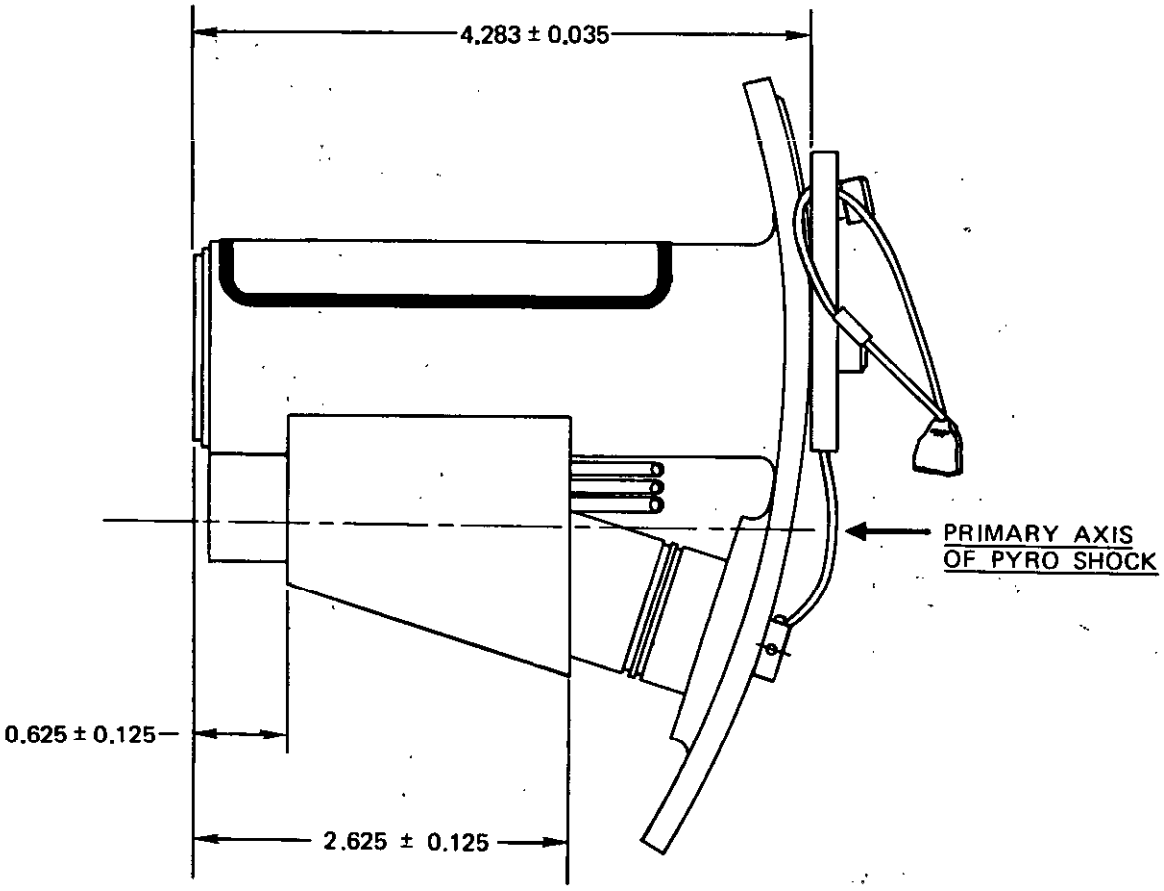


FIGURE 2. Probe configuration, type II.

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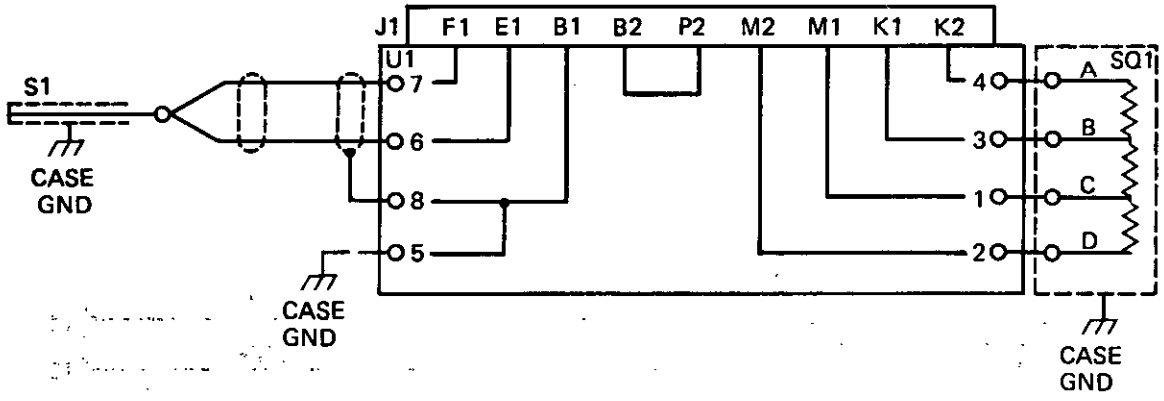


FIGURE 3. FZU-45/B, type Ia, gas generator-crush sensor electrical diagram (logic module).

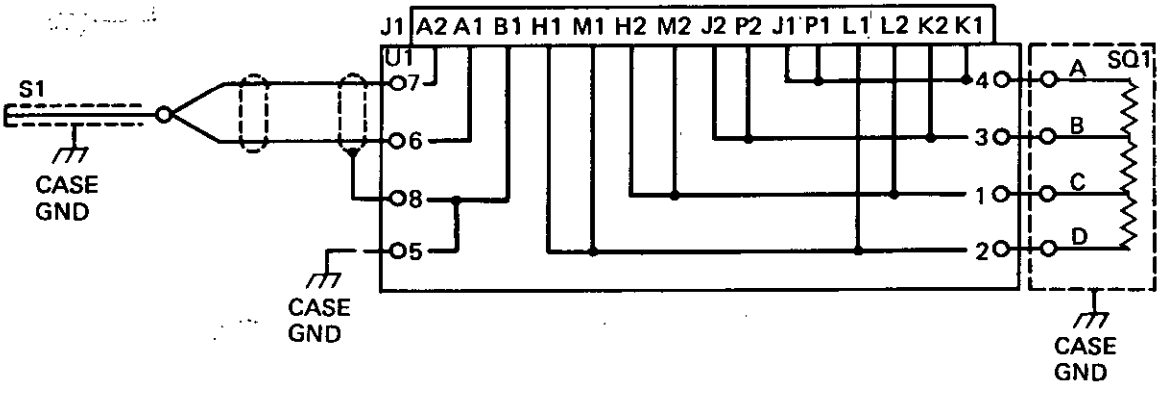


FIGURE 4. FZU-46/B, type Ib, gas generator-crush sensor electrical diagram (interface module).

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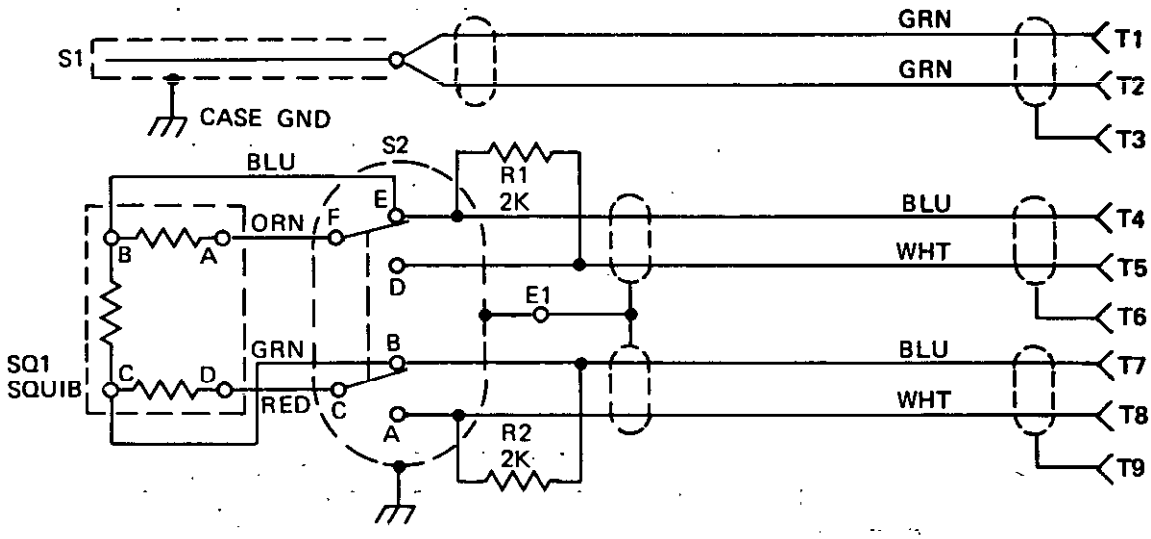


FIGURE 5. FZU-47/B, type II, gas generator-crush sensor electrical diagram.

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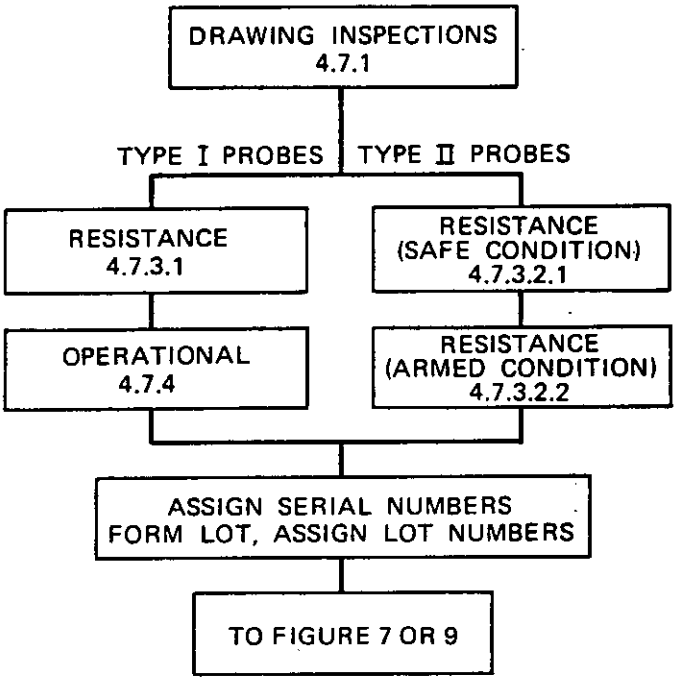


FIGURE 6. Individual unit acceptance inspection sequence.

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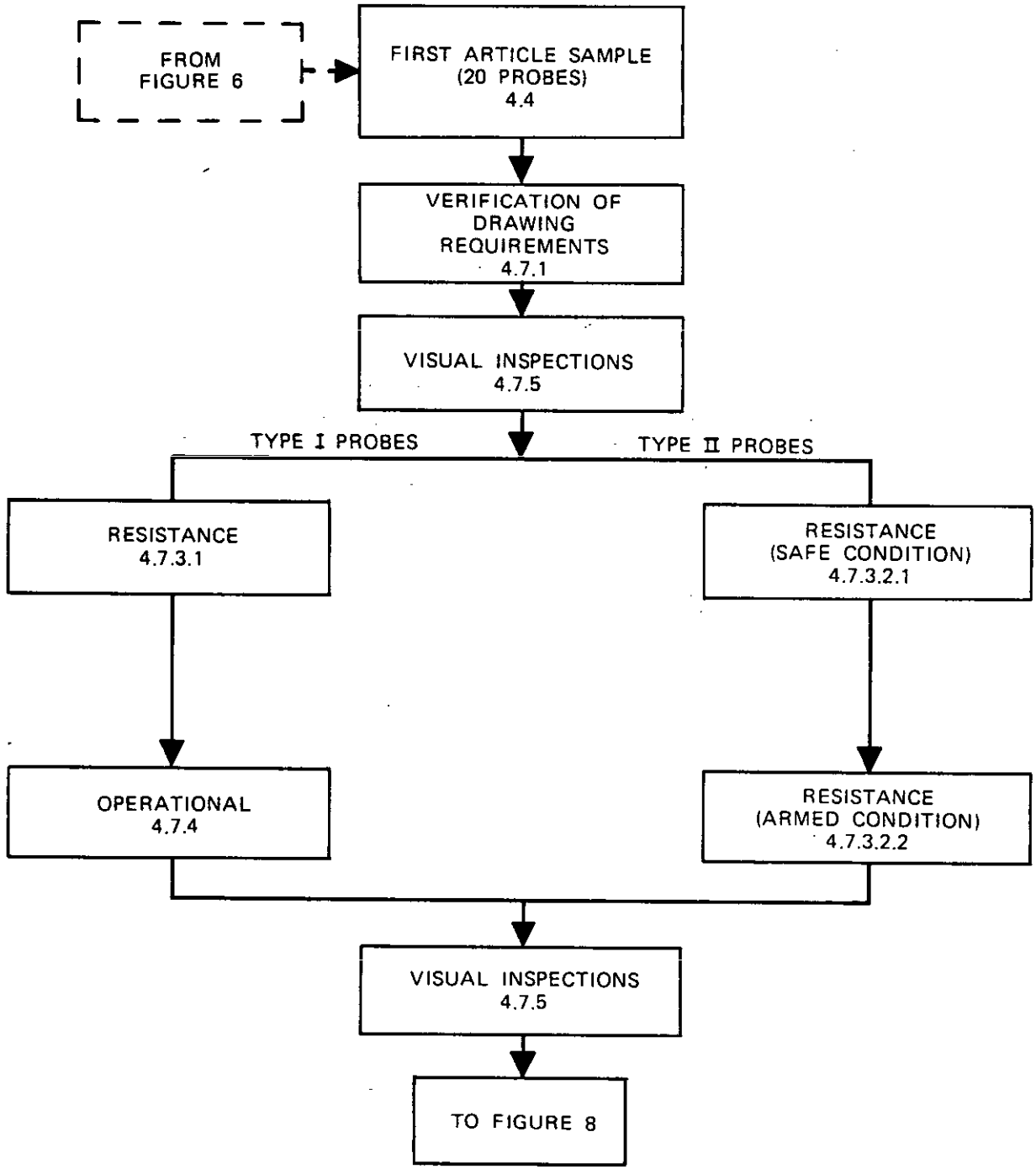


FIGURE 7. First article inspections (nondestructive).

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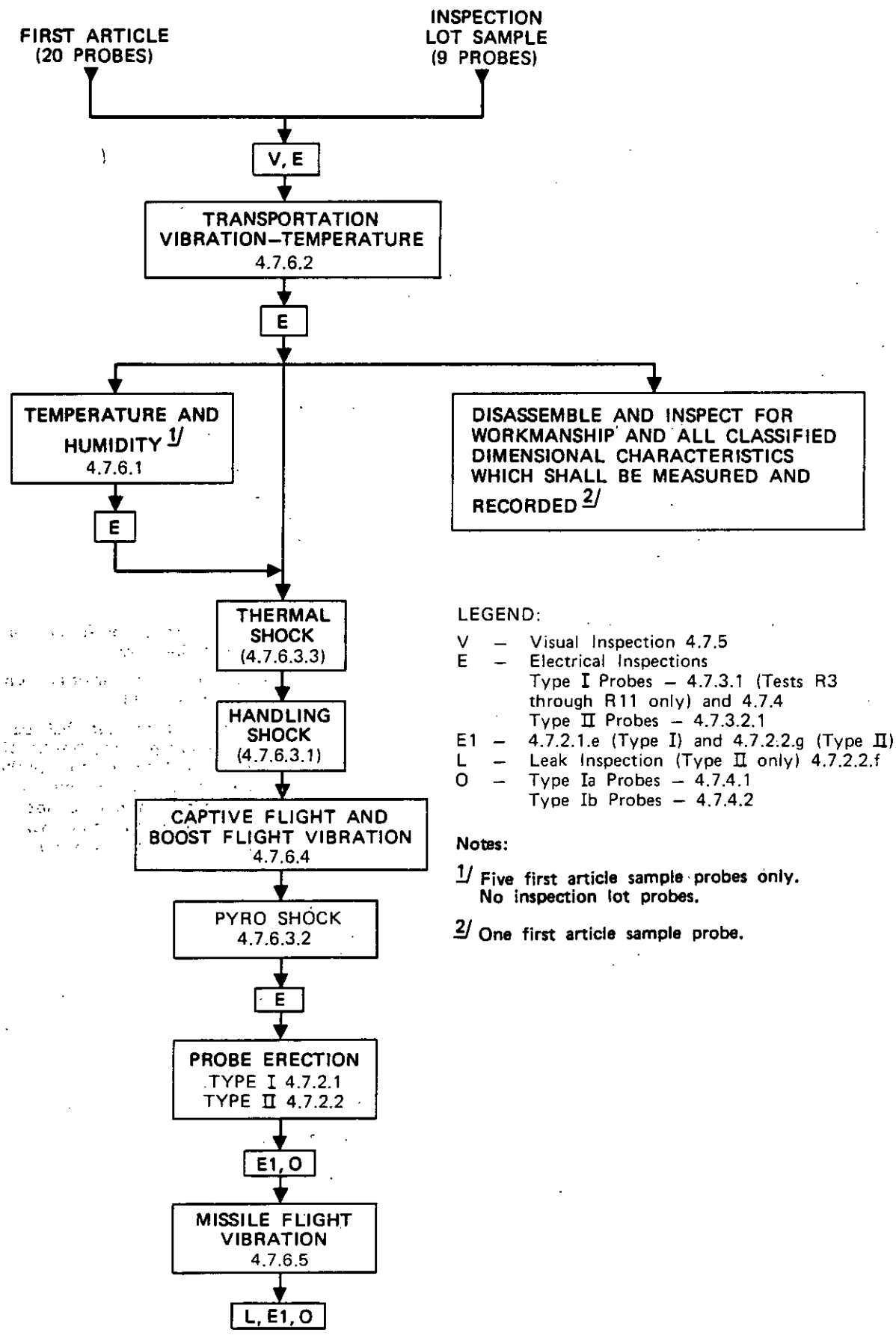


FIGURE 8. Probe environmental inspection sequence.

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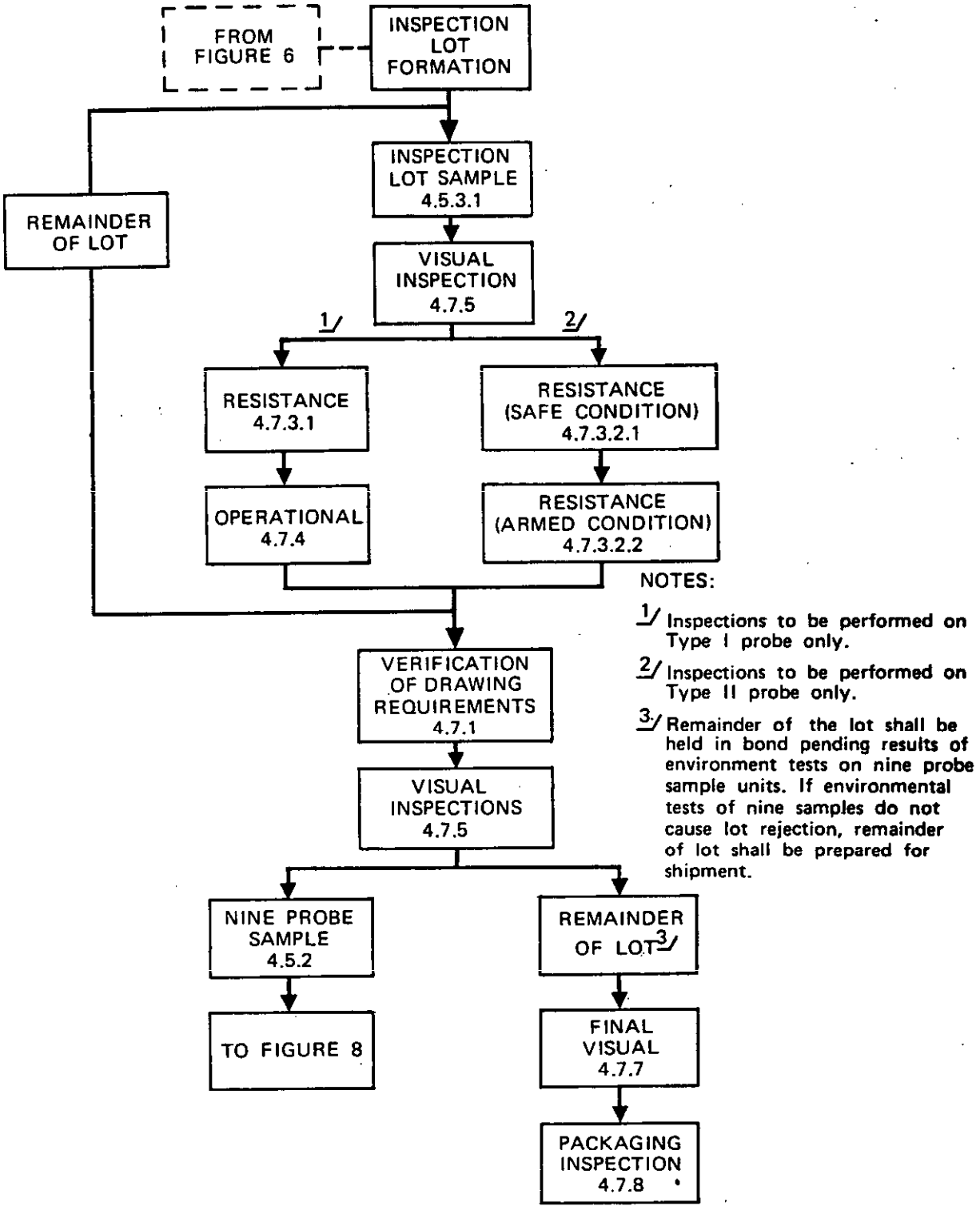


FIGURE 9. Probe lot inspection sequence (nondestructive).

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NOTES:

1. Material:
   a. Item 1: Steel, CRE, AISI type 303 or 303 SE, annealed.
      Alternate: Steel, CRE, ASTM–A–581 or ASTM–A–582 type 303 or 303 SE, condition A.
   b. Item 2: Aluminum alloy tube, 6061–T6, WW–T–700/6.

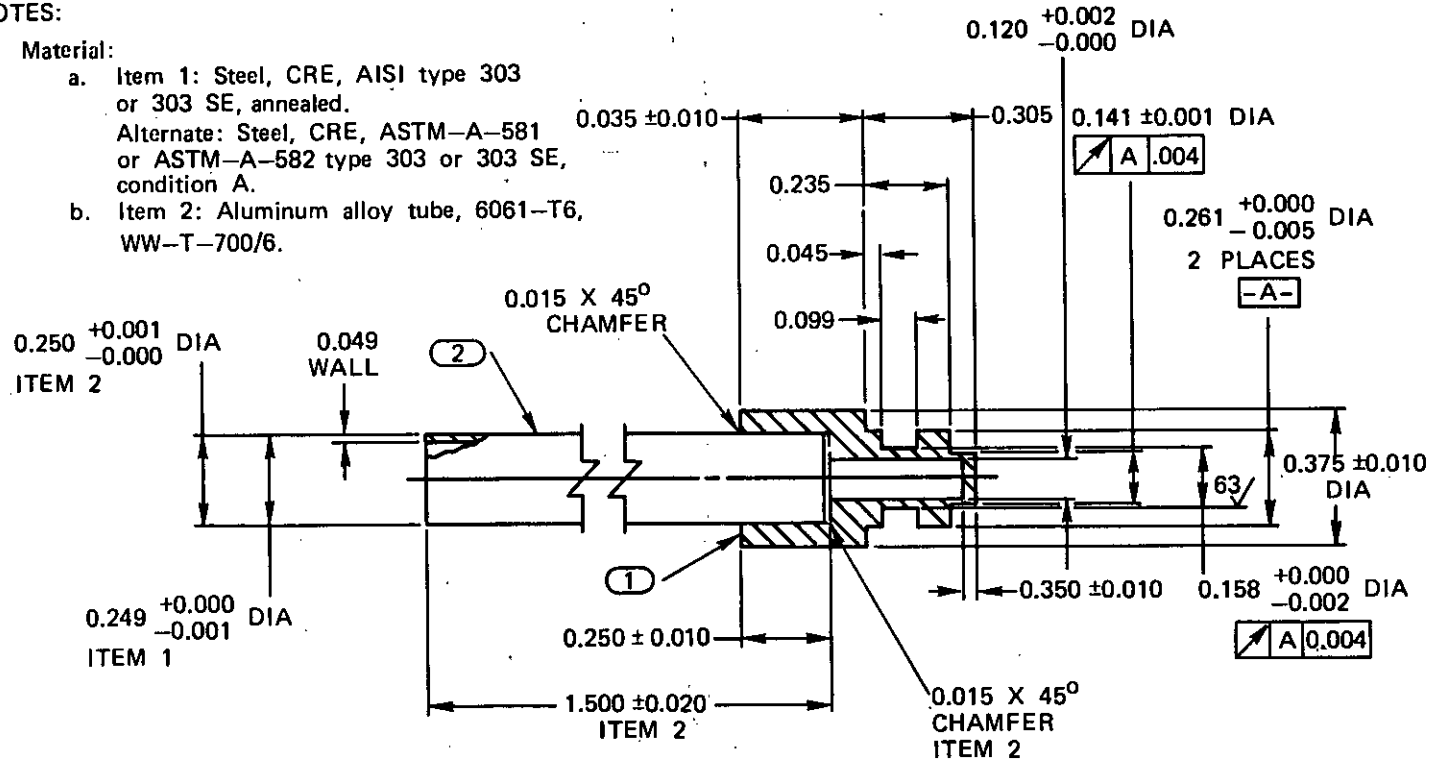


FIGURE 10. Probe simulated air tube manifold.

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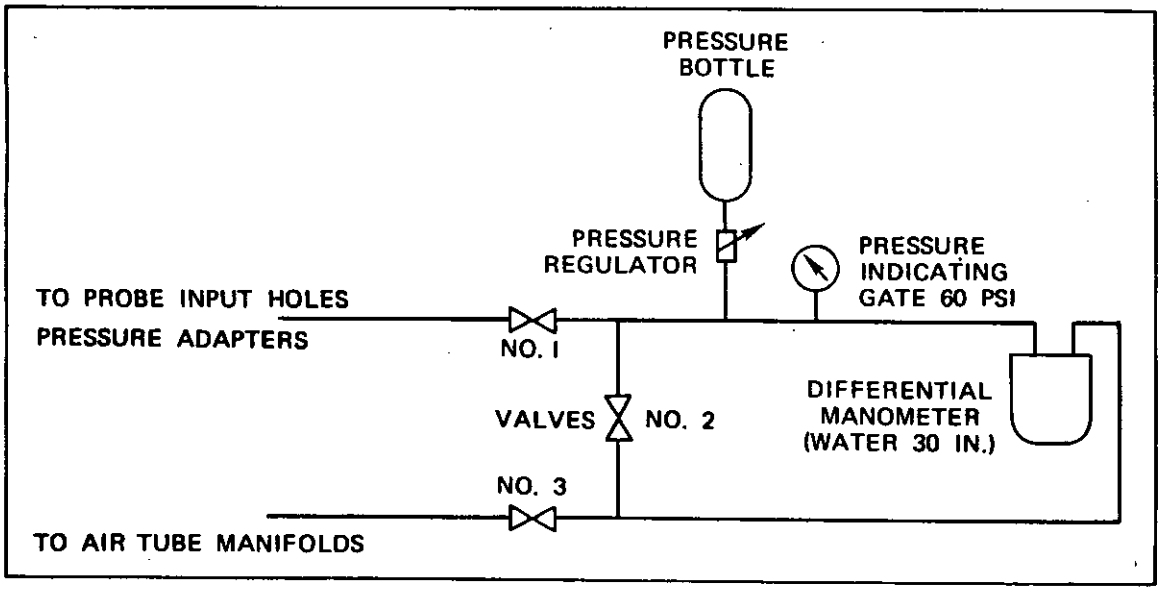


FIGURE 11. Probe pneumatic diagram for leak check.

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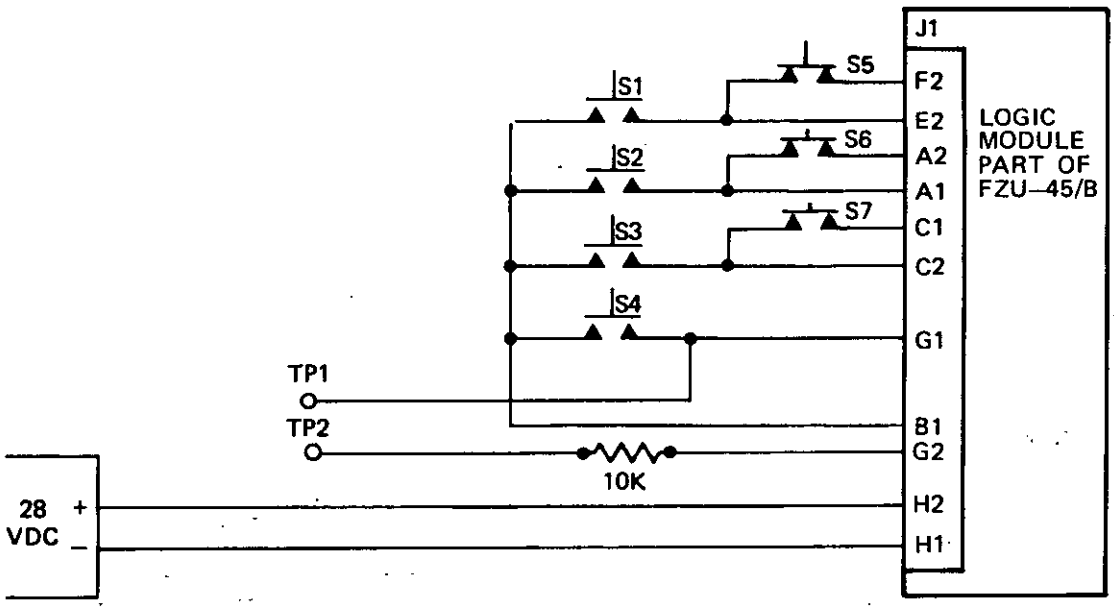


FIGURE 12. Logic module test circuit.

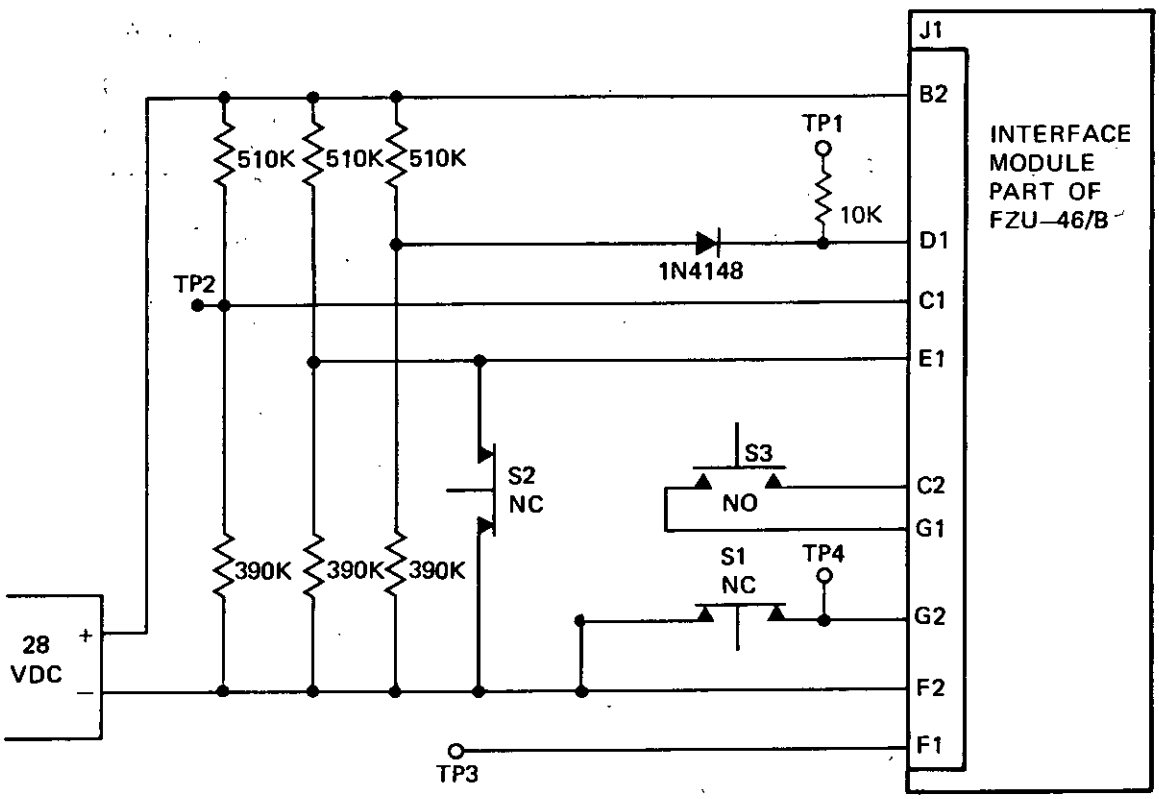
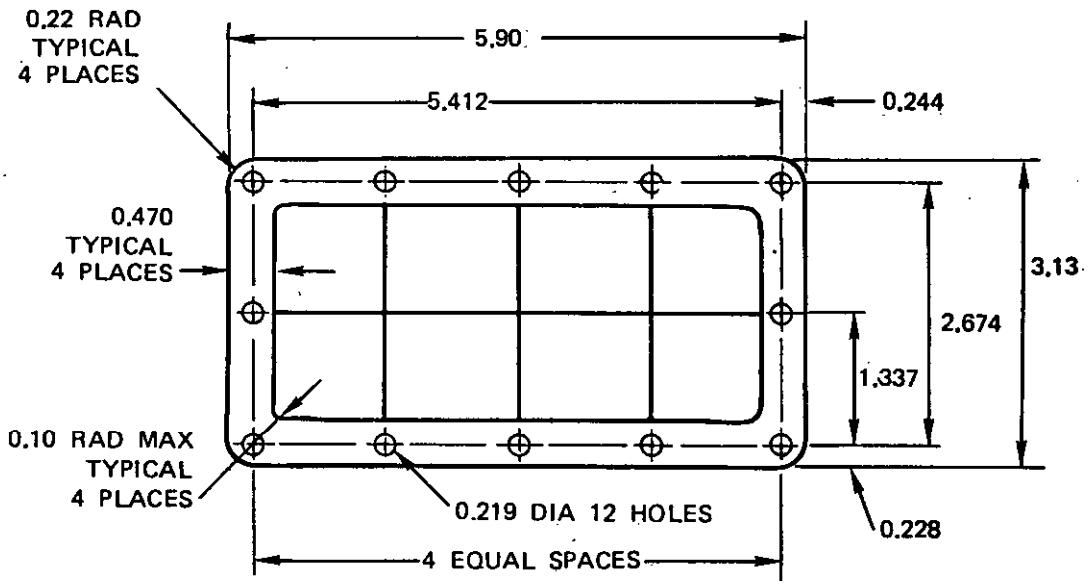


FIGURE 13. Interface module test circuit.

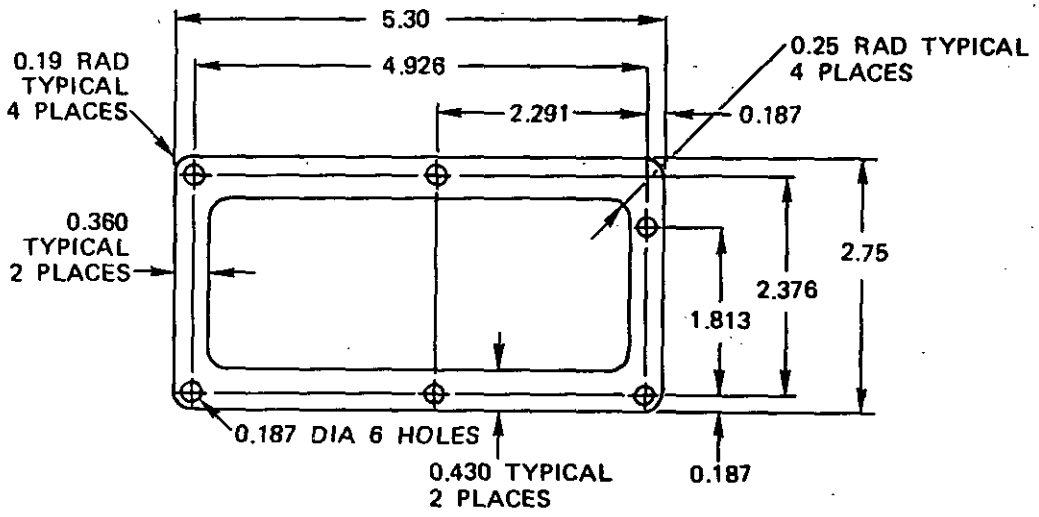
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NOTE: Material: Silicone ZZ-R-765, Class 1b, Grade 40.
Thickness: 0.062 ±0.007.

FIGURE 14. Probe gasket, type I.

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NOTE: Material: Silicone ZZ-R-765, Class 1b, Grade 40.
Thickness: 0.062 ±0.007.

FIGURE 15. Probe gasket, type II.

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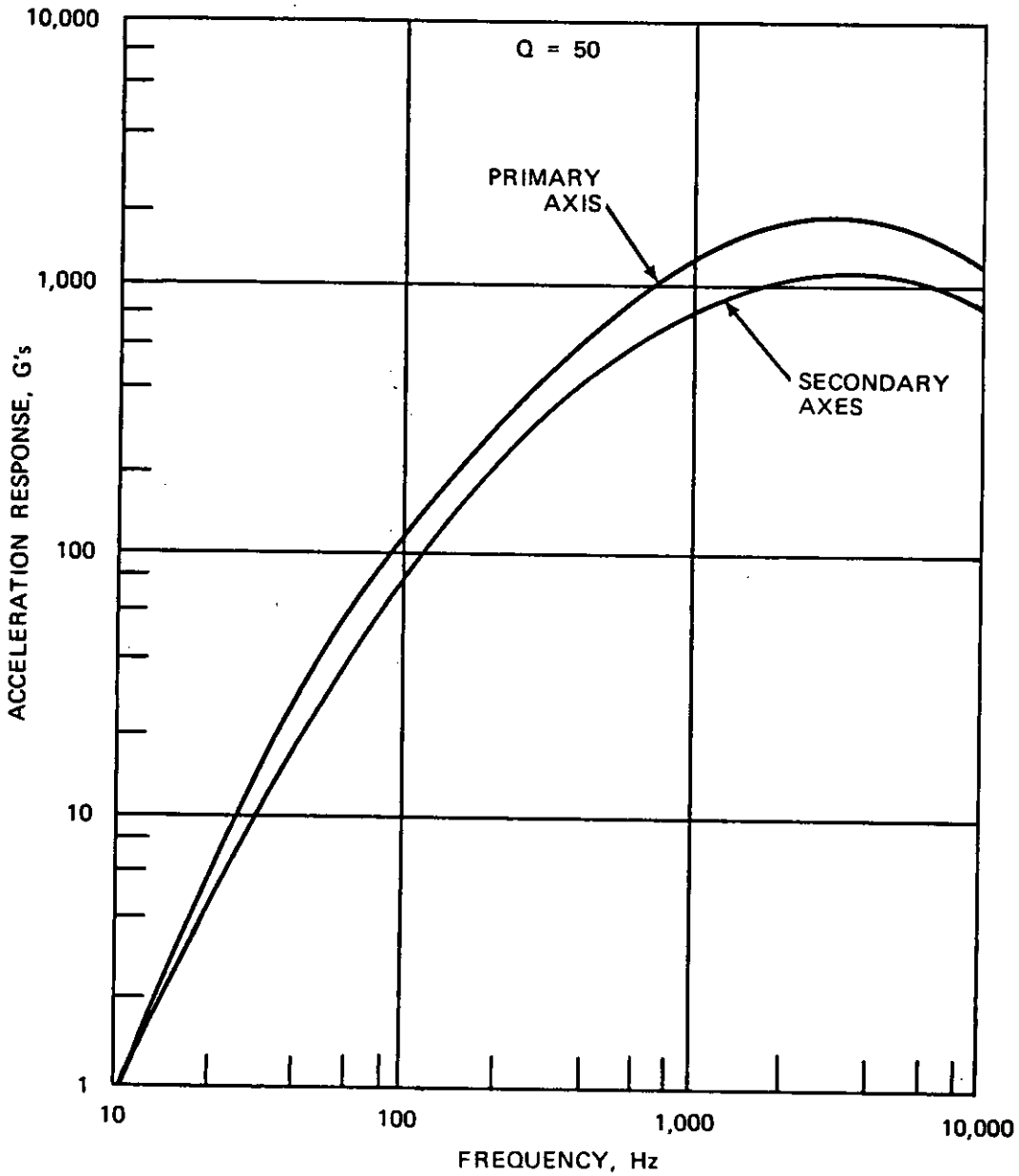


FIGURE 16. Primary and secondary pyro shock spectrum.

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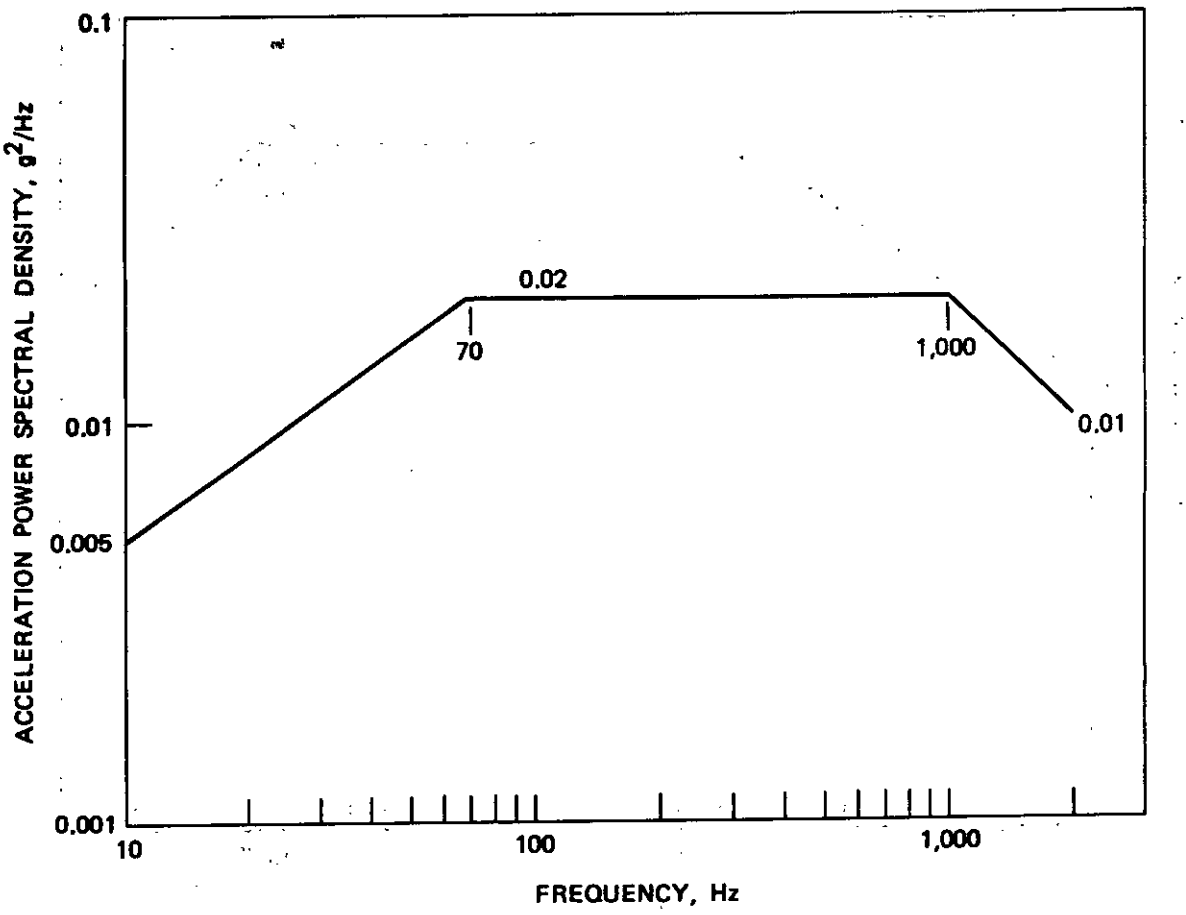


FIGURE 17. Captive and missile flight vibration.

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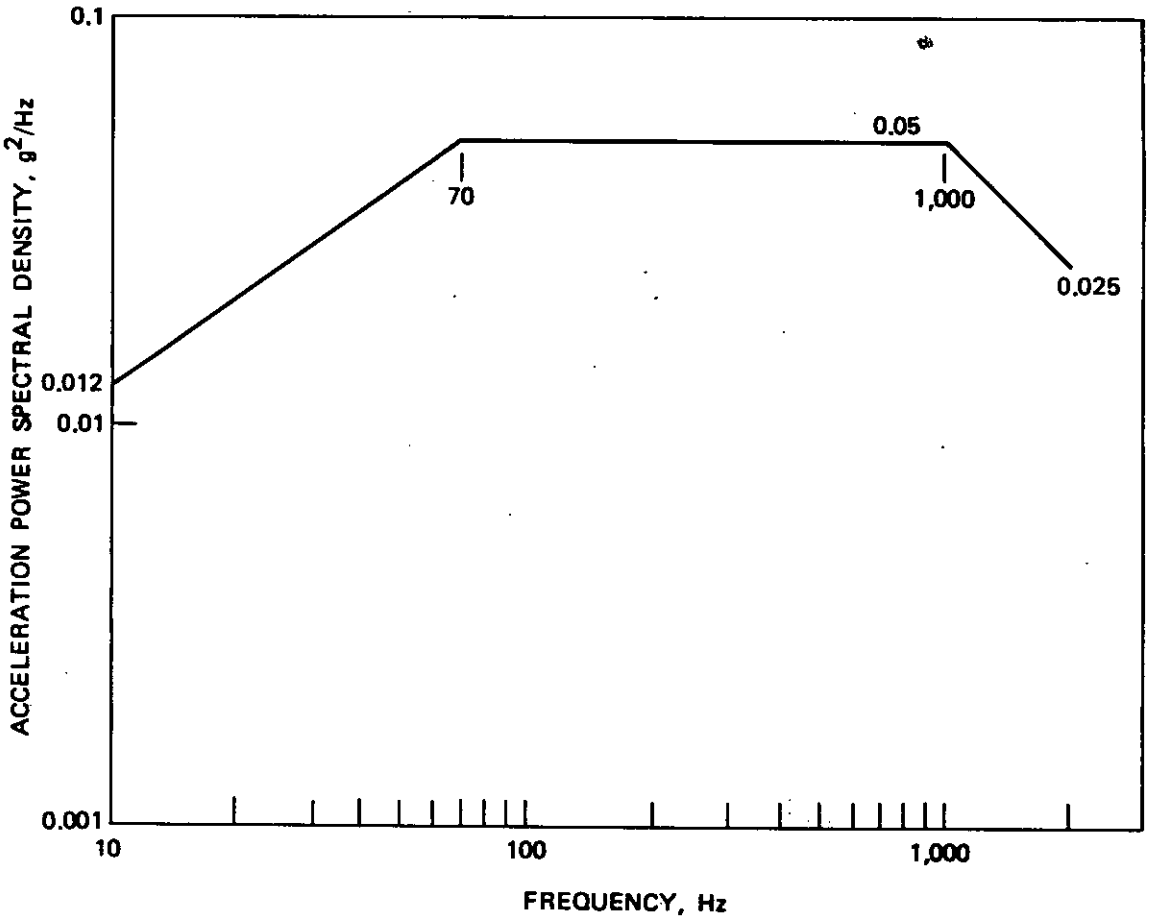


FIGURE 18. Boost flight vibration.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-85459(AS)	Probe, Crush Sensor, RH, FZU-45/B and LH, FZU-46/B and Probe, Air Pressure, FZU-47/B

3a. NAME OF SUBMITTING ORGANIZATION
Naval Weapons Center
China Lake, CA 93555

b. ADDRESS *(Street, City, State, ZIP Code)*
China Lake, CA 93555

4. TYPE OF ORGANIZATION *(Mark one)*
- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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